

LAB 02: CSS for Layout and Data Presentation

8. THINKING & DISCUSSION QUESTIONS

Question 1: Specificity War

- a) Why does nav button override .button-primary? – because nav button has higher specificity than .button-primary
- b) How would you fix this WITHOUT using !important? – use higher specificity: use nav .button-primary. Or write .button.button-primary
- c) How would you redesign the CSS to avoid this problem? – use: .button, .button-primary. And avoid tag + context such as “nav button”
- d) What does this teach you about CSS architecture? - Specificity must have a clear structure; Avoid general selectors that easily cause conflicts.

Question 2: Flexbox vs Grid

- a) Would you use Flexbox or Grid for the overall page layout? Why? – use grid because 2D layout (row + col)
- b) Would you use Flexbox or Grid for the navigation bar? Why? – use flexbox because the menu is just 1 row
- c) What if the sidebar should be on the left on desktop but move below content on mobile? – use grid + media query to change grid-template-areas
- d) Can you combine Flexbox and Grid in the same page? Give an example – yes, grid for page layout, flexbox for component

Question 3: Mobile-First vs Desktop-First

- a) What's the difference in how these work?
 - + desktop-first: design desktop first, then resize it using max-width.
 - + mobile-first: design mobile first, then resize it using min-width.
- b) What happens to a device that doesn't support media queries?
 - + desktop-first: always display layout desktop
 - + mobile-first: always display layout mobile
- c) Why is mobile-first considered better practice? – optimize performance & UX for mobile first, and CSS is easily to scale
- d) Are there cases where desktop-first makes sense? – project used for just local desktop. Or complex web applications, desktop design is the focus.

Question 4: Performance Investigation

- a) When you scroll the table, does it feel smooth or janky? – maybe a bit janky when hover many row
- b) Which CSS property is most likely causing performance issues? – box-shadow (render consumes resources)
- c) How would you optimize this for better performance? – reduce or remove box-shadow, avoid transform + shadow to multiple elements, use lighter hover effects
- d) Use DevTools Performance tab - can you see the difference? – we can see rendering time increase when hover

Question 5: CSS Variables vs Sass Variables

- a) What's the key difference between CSS variables and Sass variables? – CSS variables run runtime in the browser. Sass variables compile-time when building CSS
- b) What can CSS variables do that Sass variables cannot? – dynamic change (ex: dark mode)
- c) What can Sass variables do that CSS variables cannot? – use in logic build such as loops, functions
- d) When would you use each?
 - + CSS var: themes, dynamic styles
 - + Sass var: manage design system when building css

Question 6: The Cascade in Action

- a) What color is the text? Why? – green, because #special has the highest specificity
- b) If you remove the id, what color would it be? – red, because p.intro more specify than .intro
- c) What if there's also an inline style: <p style="color: yellow">? – yellow, because inline style > all css selector
- d) How would you calculate specificity for p.intro.featured#special? – (1,2,1) → 1 id, 2 class, 1 element

Question 7: Real-World Color Systems

- a) How would you organize these colors using CSS variables? –declare in :root such as: --color-primary; --color-;...
- b) How would you create lighter/darker versions without design tools? – use filter, hsl(), or color-mix()
- c) What about transparency variations? – use rgba()
- d) Design a system where changing ONE variable updates the entire color scheme.
 - use theme variables: change --color-primary will update all component.

Question 8: Debugging CSS Layout Issues

- a) Why do boxes overflow instead of wrapping? – because default wrap is no wrap (flex-wrap: nowrap)
- b) What's the minimum CSS change to fix this? – flex-wrap: wrap
- c) How would you use DevTools to diagnose this? – inspect .container → inspect the flex overlay and computed width.
- d) What if you wanted boxes to shrink instead of wrap? – use flex: 1

Question 9: Accessibility vs Aesthetics

- a) How would you fix accessibility issues while keeping the design appealing? – increase contrast, add :focus outline, increase font > 16px
- b) What compromises would you make? – keep the gradient but darken the background or lighten the text.
- c) How do you have this conversation with a designer? – explain WCAG requirement + UX for visually impaired people.
- d) What if the client insists on the original design? – higher risk in document: accessibility/legal issues

Question 10: CSS in Large-Scale Applications

- a) What problems would you expect to encounter? – css conflict, harder to maintain, bug layout
- b) How would you propose refactoring this? - Break it down into smaller modules/components.
- c) What conventions/tools would you introduce? – use BEM naming, CSS variables, stylelint, documentation
- d) How do you prevent this from happening again? – code review + design system
- e) At what point would you consider a CSS-in-JS solution? – when app React is bigger and need component-scoped styles